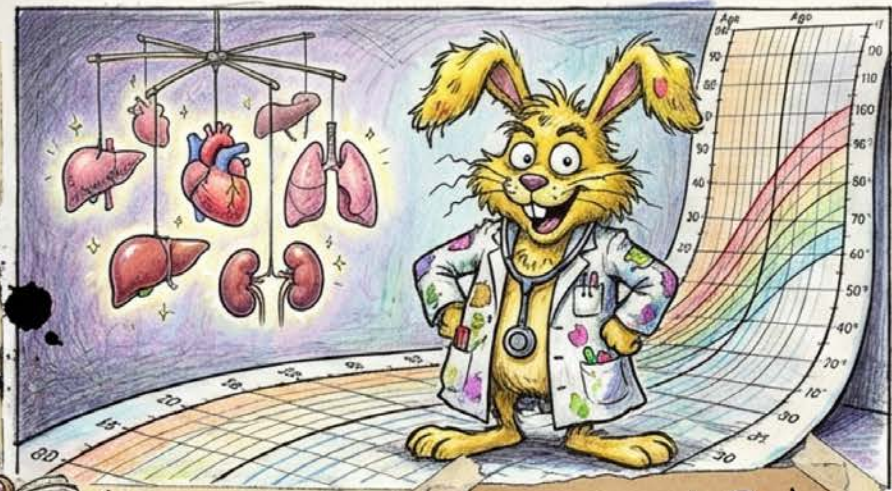


Georg von Westphalen

PEDIATRICS: THE GROWING GAUNTLET

Protecting the future,
one percentile at a time.



Doc FrizzleBob

You remember stories, not lists.
Every mechanism is a character.
Every disease is a wrestling move.

Welcome to the stage of the body.



BLÖDSINN!

Children are NOT small adults.

They are dynamic biological systems.

Your job is to navigate the Growth Chart

Crossroads and defend the Immune-Innocents.

Stay fluffy!

Call Doc Frizzle!



Apgar-the-Scribe (The First Score)



STAT: Appearance, Pulse, Grimace, Activity, Respiration (Score 0-10).

POWER (The Welcome-Check): Judges the Newborn Knight at 1 and 5 mins for Resuscitation-Rescue.

LORE: Score < 7 is a call for backup. Reports only on the Transition-Triumph from the womb.

Surfactant-the-Slippery (RDS)



STAT: Produced by Type II Pneumocytes / Lecithin-Sphingomyelin ratio.

POWER (The Anti-Collapse): Reduces tension in Alveolar-Balloons, keeping them open.

LORE: Without it, lungs turn to Stiff-Stone (Hyaline Membrane). Give Steroid Stella to mom.

KayfaBONGO!

Bilirubin-the-Yellow (Jaundice)



STAT: Indirect vs. Direct Phototherapy threshold Kernicterus risk.

POWER (The Skin-Stainer): Liver-Factory is too slow or Red-Cell-Inventory breaks too fast.

LORE: Yellow-Mist crosses the blood-brain barrier into the Basal-Ganglia-Bunker (Kernicterus). Blue light fixes it.

Duct-the-Diverter (PDA)



STAT: Machinery Murmur PGE keeps it open Indomethacin closes it.

POWER (The Fetal-Shortcut): Bypasses lungs in the womb, must Snap-Shut after the first breath.

LORE: Use Indo-Lock to close, or Prosta-Plug to keep open for Cyanotic-Secrets.



VSD-the-Vortex (Ventricular Septal Defect)



STAT: Holosystolic murmur / Left-to-Right shunt.

POWER (The Wall-Leak): Oxygenated blood spills back into the Lung-Pump.

LORE: Most common defect. Pressure rises until the Shunt-Flips (Eisenmenger-the-End-Game).

Tetralogy-the-Tet (Cyanotic Heart)



STAT: PROV: Pulm stenosis, RV hypertrophy, Overriding aorta, VSD.

POWER (The Blue-Spell): Crying makes them turn blue as blood bypasses lungs.

LORE: The Squat-Move increases systemic resistance, forcing blood back to lungs.

Coarct-the-Kink (Aortic Coarctation)

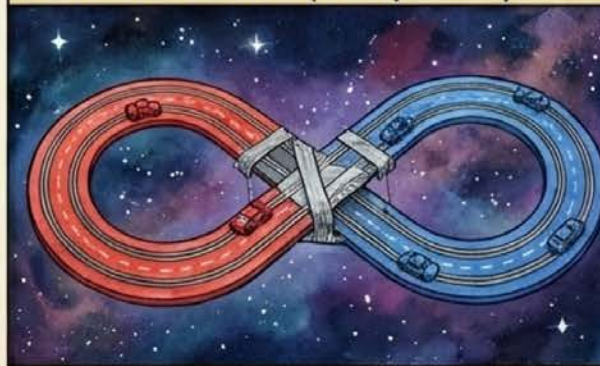


STAT: BP higher in arms than legs / Weak femoral pulses / Rib notching.

POWER (The Pressure-Split): Narrowing in the Main-Highway stops flow to Lower-Limb-Lands.

LORE: Turner-the-Tough association. Heart pumps like a maniac to get past the Kink.

TGA-the-Switch (Transposition)



STAT: Egg on a string X-ray / Aorta and PA are swapped.

POWER (The Parallel-Paradox): Lungs and body are in two separate circles.

LORE: Lethal without a Mixing-Hole. A Keep-it-Open-Emergency for Duct-the-Diverter.

Diagnosis?!



RSV-the-Wheezer (Bronchiolitis)



STAT: Winter season
Infants < 2 years
Retractions

POWER (The Mucus-Plunger): Inflames tunnels, making every breath a struggle.

LORE: Not a pneumonia, a tunnel blockage. Suction and hydration only.

Croup-the-Barker (Laryngotracheobronchitis)



STAT: Parainfluenza virus / Barking Seal cough / Steeple Sign

POWER (The Subglottic-Squeeze): Narrows the pipe below Vocal-Vibrators (Stridor-Whistle).

LORE: Loves cold night air. Give Dexa-the-Dot (Steroids) to calm the squeeze.

What the FLUFF?!

Pertussis-the-Whoop (Whooping Cough)



STAT: Paroxysmal cough
Inspiratory whoop
Lymphocytosis

POWER (The Machine-Gun-Cough): Violent fits ending in a gasping, high-pitched whoop.

LORE: Bordetella paralyzes the Cilia-Cleaners, locking mucus in place.

Kawasaki-the-King (Vasculitis)



STAT: CRASH & Burn
Conjunctivitis, Rash, Adenopathy, Strawberry tongue, Hands/feet swelling + Fever > 5d.

POWER (The Heart-Coronary-Threat): Inflames Heart-Pipes into Aneurysm-Balloons.

LORE: Needs Aspirin-the-Ace and IVIG-the-Shield. The ONE time you give kids aspirin!

13 Pyloric-the-Projector (Stenosis)

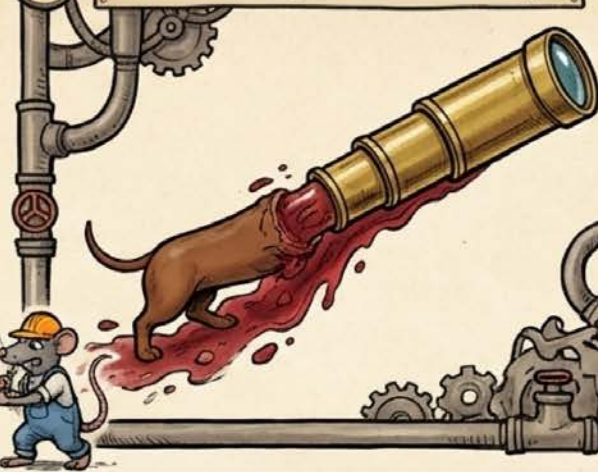


STAT: 3-6 weeks old
Olive mass
Non-bilious
projectile vomiting.

POWER (The Muscle-Wall):
Overgrown muscle blocks
all Milk-Traffic.

LORE: Causes
Hypochloremic Alkalosis.
Fis Salt-and-Acid-Stats
before the Blade-Brigade
cuts the muscle.

14 Intussuscept-the-Telescope



STAT: 6-36 months
Currant Jelly stool
Sausage-shaped
mass.

POWER (The Slide-In):
Bowel telescopes, cutting
off Blood-Supply-Road.

LORE: Sudden
screaming pain. The
Air-Enema-Push
blows the telescope
back out.

15 Hirschsprung-the-Static (Megacolon)



STAT: Delayed
meconium (>48h)
Squirt sign
Aganglionic segment

POWER (The Nerve-Gap):
Missing Neural-Drivers,
stays static and never
squeezes.

LORE: Everything
upstream swells into a
Mega-Colon.

16 Meckel-the-Mimic (Diverticulum)



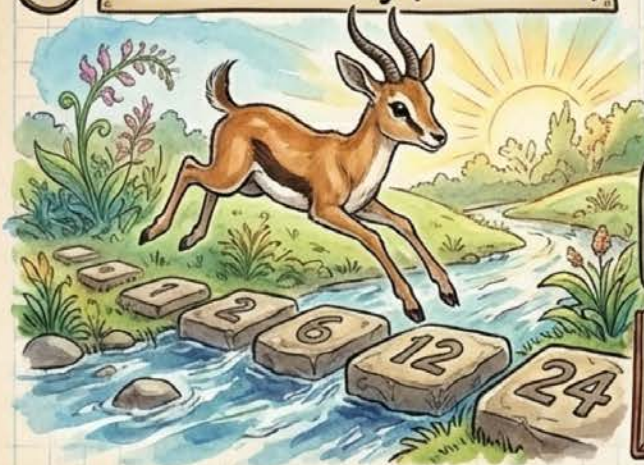
STAT: Painless
rectal bleeding
Rule of 2s (2%, 2
inches, 2 feet from
valve)

POWER (The Hidden-Bleeder):
Leftover pouch with
Stomach-Acid-Cells
burns the bowel.

LORE: Diagnosed by
Meckel-Scan (Technetium-99m) to find the
acid cells.

KayfaBINGO!

17 Milestone-Molly (The Tracker)



STAT: Social smile (2m), Sit (6m), Walk (12m), 2-word sentences (2y).

POWER (The Growth-Gage): Tracks the Skill-Inventory.

LORE: Loss of a milestone is a Red-Flag-Emergency (Regression-Riot).

19 ADHD-the-Impulse (Alex)



STAT: Inattention, Hyperactivity, Impulsivity / Two settings (Home/School).

POWER (The Squirrel-Chase): A hundred drones buzzing in the ring block focus.

LORE: Low Dopamine-Dust. Stimulant-Save helps turn the drones off.

18 Autism-the-Alchemist



STAT: Social communication deficits / Restricted, repetitive patterns.

POWER (The Different-Frequency): Processes the world through a unique filter, missing Social-Signal-Pings.

LORE: Early intervention is the Transmuter via Speech-Spells and Social-Scripts.

20 Failure-the-Thrive (FTT)



STAT: Weight drops below 3rd percentile / Crosses 2 major percentiles down.

POWER (The Energy-Drain): Not getting enough Fuel-Tokens or burning them too fast.

LORE: Organic vs Non-organic. The Growth-Chart is the first witness.

MedKayfabe!

21 Febrile-the-Flash (Seizure)



STAT: 6 months to 5 years / Simple (<15 min) vs. Complex.

POWER (The Temperature-Short): Rapid heat rise causes a brain Glitch.

LORE: Benign but scary. Just a System-Reset, does not damage the brain.

22 Epiglottitis-Emma (The Choke-Point)



STAT: Drooling, Tripoding, Stridor, Fever / Thumbprint on X-ray.

POWER (The Airway-Slam-Door): High-priority assassin shuts the respiratory runway in seconds.

LORE: Keep the kid calm! Agitation triggers Final-Closure. Prep the Tube-Tango.

23 Meningitis-the-Menace (Neonatal)



STAT: Bulging fontanelle / Lethargy / GBS, E. coli, Listeria.

POWER (The Brain-Shield-Riot): Attacks the Meninges-Armor.

LORE: Neonates don't get Stiff-Necks, they get Limp-Bodies and High-Pitched-Screams.

DANGER!

24 SIDS-the-Silent (Sudden Infant Death)



STAT: Peak 2-4 months / Back to Sleep prevention.

POWER (The Quiet-Fade): Sudden, unexplained end to the match.

LORE: Prevented by Safe-Sleep-Spells: No blankets, back-sleeping, smoke-free.



KayfaBINGO!

GW

Georg v. Westphalen

DANGER!

25 Pancreas-Factory (The Cryster)



STAT: Polyuria,
Polydipsia,
Weight loss /
DKA risk.

POWER (The Insulin-Strike): Factory destroyed by Self-Attack-Revolt.

LORE: High sugar =
DKA-Acidosis-Abyss.
Needs Insu-the-Key
forever.

26 Beaver-the-Wolk (Salt-Savers)



STAT: 21-hydroxylase
deficiency /
Low Na, High K /
Ambiguous genitalia.

POWER (The Hormone-Shortcut): Cannot make
salt/sugar hormones,
defaults to androgens.

LORE: Electrolyte
emergency. Without
salt-savers, the
wrestler collapses.

27 Friendly-the-Retrivor (Wrestler)



STAT: Hypotonia,
Simian crease,
Endocardial
cushion defects.

POWER (The Extra-Blueprint): Extra
Chromosome-21 changes
the Physiological-Rulebook.

LORE: High risk for
Heart-Leaks and Gut-
Blocks. Maximize potential
with Support-Squads.

28 Sloth-the-Sludge (Bio Wrestler)



STAT: Sweat chloride
test / CFTR mutation
/ Pancreatic
insufficiency.

POWER (The Sticky-Sludge): Broken
Salt-Pumps turn
secretions into glue.

LORE: Causes Chronic-
Moni-Pneumonia and
Fatty-Stool-Steve. Needs
Chest-Physio-Pounding.

MedKayfabe!

KayfaBINGO!

29 Nursemaid-Nancy (Radial Head Subluxation)



STAT: Sudden pull on arm / Refuses to use arm / Held in pronation.

POWER (The Annular-Tear): Ligament caught inside elbow-gear after a Sudden-Lift.

LORE: The Supination-Flexion-Flip pops it out, curing the wrestler instantly.

30 SCFE-Scottie (Slipped Capital Femoral Epiphysis)



STAT: Obese adolescent / Knee pain Ice cream falling off the cone.

POWER (The Sliding-Hip-Ball): Hip-ball slides off the Growth-Plate-Ramp.

LORE: Knee pain is often referred hip pain. Surgical emergency to prevent Hip-Death.

31 Atopic-Alice (Eczema)



STAT: The Itch that Rashes / Flexural surfaces / Filaggrin defect.

POWER (The Leaky-Skin-Wall): Skin-armor holes let moisture out and invaders in.

LORE: Part of the Atopic-Triad. Needs heavy Moisture-Maintenance.

32 Measles-the-Miser (Rubeola)



STAT: 4 Cs: Cough, Coryza, Conjunctivitis, Coplik spots.

POWER (The Total-Body-Exanthem): Starts with red eyes, covers body in Sandpaper-Rash.

LORE: The Immune-Amnesia virus. Wipes out Memory-B-Cells.

KayfaBOGGLE!

KayfaBINGO!

33 Tanner-the-Stage (Puberty)

STAT: Stages 1-5
(Pubic hair,
Breast/Genital
development).



POWER (The Metamorphosis):
Tracks the Hormonal-Upgrade
to adult form.

LORE:
Precocious-Pete starts early,
Delayed-Dave stays behind.

34 Scolio-the-S-Curve

STAT: Cobb Angle
> 10 degrees /
Adams Forward
Bend Test.



POWER (The Spine-Twist):
Turns straight Skeletal-Pillar
into a snake.

LORE: Mostly idiopathic.
Severe curves squash the
Lung-Balloons.

35 Anorexia-Annie (Eating Disorder)

STAT: BMI < 17.5 /
Body dysmorphic
disorder / Lanugo.



POWER (The Mirror-Lie):
Sees a giant despite
being skin and bones.

LORE: Danger: Refeeding-Ryan-the-
Relapse. Eating too fast crashes
electrolytes and the heart.

36 Suicidal-the-Shadow (Adolescent Mental Health)

STAT: Second leading
cause of death /
Depression screening.



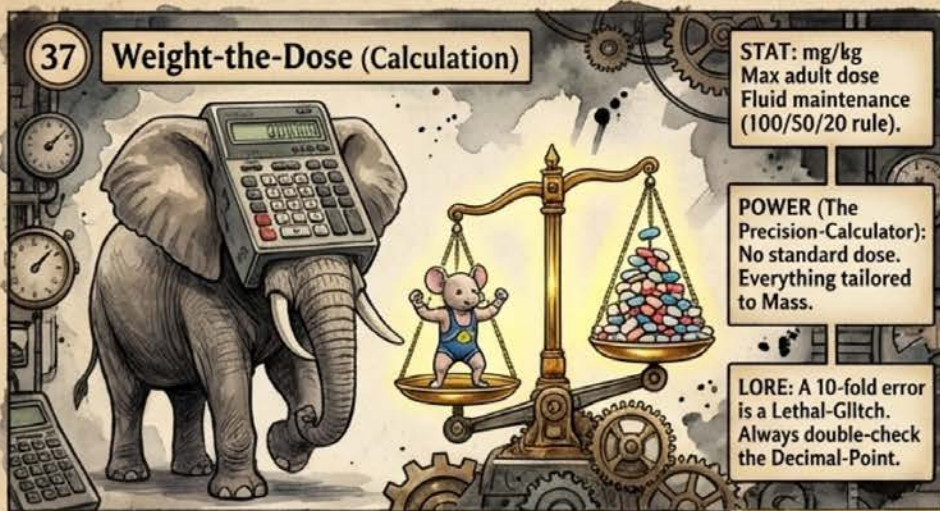
POWER (The Silent-Saboteur):
Whispers that the match
isn't worth finishing.

LORE: Always ask Safety-Questions.
Look for Bullying-Brawlers
and Isolation-Iggles.



37

Weight-the-Dose (Calculation)



STAT: mg/kg
Max adult dose
Fluid maintenance
(100/50/20 rule).

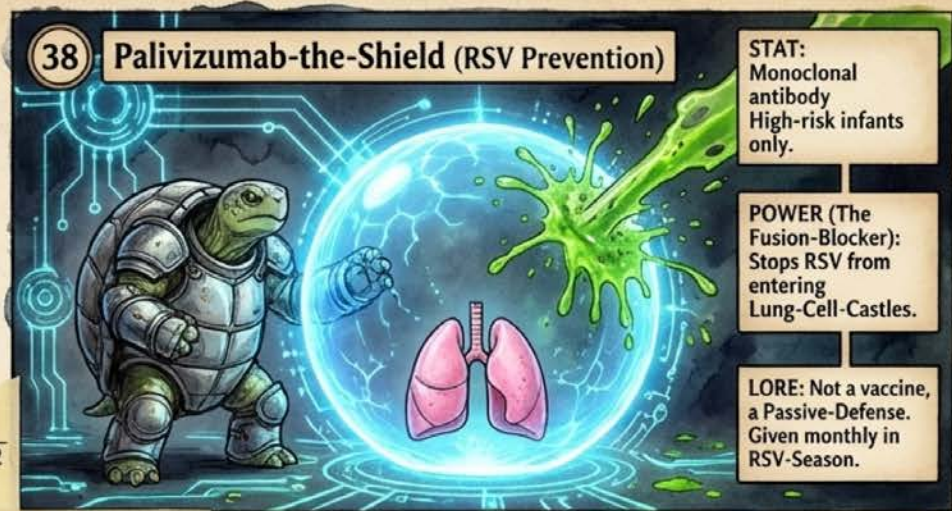
POWER (The Precision-Calculator):
No standard dose.
Everything tailored to Mass.

LORE: A 10-fold error
is a Lethal-Glitch.
Always double-check
the Decimal-Point.

$$\frac{-1X}{V2}$$

38

Palivizumab-the-Shield (RSV Prevention)



STAT:
Monoclonal
antibody
High-risk infants
only.

POWER (The Fusion-Blocker):
Stops RSV from
entering
Lung-Cell-Castles.

LORE: Not a vaccine,
a Passive-Defense.
Given monthly in
RSV-Season.

39

Steroid-Stella (Antenatal)



STAT: Betamethasone
/ Dexamethasone
Before 34 weeks.

POWER (The Lung-Accelerator):
Speeds up the
Surfactant-Factory.

LORE: A 48-hour
Turbo-Boost to
prep the Premie-
Wrestler for air.

$$\frac{-1X}{V2}$$

40

Acetamino-the-Fever-Cooler



STAT: 10-15 mg/kg
Avoid Aspirin
(Reye's Syndrome).

POWER (The Heat-Sink):
Turns down the
Brain-Thermostat
during the Flash-Fever.

LORE: NO ASPIRIN
FOR KIDS! Triggers
Reye-the-Liver-and-
Brain-Smasher.

41 The MMR-Trio (Live Vaccine)



Measles, Mumps, Rubella /
12 months and 4 years.

POWER (The Training-Sim):
Provides a practice match
against three classic
assassins.

LORE: Live-attenuated.
NOT for Immune-Depleted
or pregnant wrestlers.



42 DTaP-the-Triple-Threat



STAT: Diphtheria, Tetanus,
acellular Pertussis.

POWER (The Toxin-Shield):
Blocks Strangling-
Diphtheria, Lockjaw-
Tetanus, and
Whooping-Pertussis.

LORE: Needs boosters
every 10 years to keep
Defense-Towers standing.

Stay fluffy!

43 Polio-the-Paralyzer (IPV)



STAT: Salk (Inactivated) /
Eradication goal.

POWER (The Nerve-
Shield): Protects Motor-
Neurons from being
hijacked.

LORE: We use IPV (Dead)
to avoid the rare Vaccine-
Induced-Paralysis of the
old live version.

44 Hib-the-B-Shield



STAT: Haemophilus
influenzae type b.

POWER (The
Brain-and-Throat-Guard):
Eliminated
Emma-Epiglottis and
Meningitis-Menace.

LORE: Conjugate vaccine
teaches cells to read the
Bacteria's-Sugar-Coat.

45 ALL-the-Acute-Lymphoblastic



STAT: Most common childhood cancer
Bone marrow blast > 20% TDT+.

POWER (The Marrow-Invasion): Blast-Punks stop production of inventory and Platelet-Plugs.

LORE: Highly curable but needs years of Chemo-Combat. Look for bone ache and bleeding spots.

46 Wilms-the-Kidney-Wrestler



STAT: Nephroblastoma
Does NOT cross the midline
Smooth mass.

POWER (The Flank-Fist): Massive tumor inside the Kidney-Filter.

LORE: Do Not Palpate! Squeezing pops it, spreading Seed-Tumors into the belly.

BLÖDSINN!

47 Neuro-the-Crosser (Neuroblastoma)



STAT: Adrenal origin
Crosses the midline
Calcifications

POWER (The Midnight-Mass): Irregular tumor spreading everywhere from the basement.

LORE: Associated with Dancing-Eyes-Dancing-Feet (Opsoclonus-Myoclonus).

48 Retino-the-White-Reflex (Retinoblastoma)



STAT: Leukocoria (White pupil)
RB1 gene mutation

POWER (The Eye-Shadow): Tumor on the Retinal-Film-Strip blocks the red reflex.

LORE: Often genetic. The Cat's-Eye-Glow in a photo is the ultimate Diagnostic-Red-Flag.

49 The Growth-Chart-Oracle

STAT: Weight-for-age,
Height-for-age,
Head circumference.

POWER (The Pattern-Predictor): Tells if the wrestler is falling behind or growing too fast.

LORE: The Trend is everything. One point is a dot; two points are a story.



50 The Red-Reflex Sentry

STAT: Ophthalmoscope exam in a dark room.

POWER (The Vision-Guardian): Checks if light can reach the back of the eye.

LORE: If the reflex is white (Leuko-Leo) or absent, the Airway-to-Vision is blocked.



51 The Sweat-Test Detective

STAT: Pilocarpine iontophoresis / $\text{Cl}^- > 60 \text{ mmol/L}$.

POWER (The Salt-Seeker): Hunts for chloride.

LORE: If the sweat is Salty-as-the-Sea, the Cystic-Clog is the culprit.

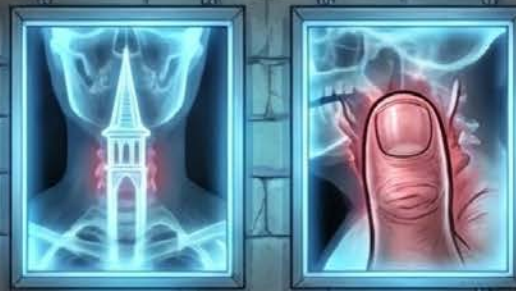


52 The Steeple-and-Thumb Signs

STAT: Lateral neck vs. AP Chest X-rays.

POWER (The Airway-Map): Immediate visual triage of the airway.

LORE: Steeple = Croup-Chris (Safe-ish).
Thumb = Epiglottis-Emma (Danger!).





The Pediatric Attending (Doc FrizzleBob)

STAT: Board Certified /
Specialist in Small Humans.

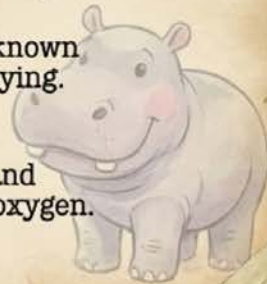
POWER (The Future-Shield):
The ultimate defender of the
growing gauntlet.

LORE: Are milestones met?
Is the airway safe? Check the
sugar, check the chart, and
don't forget the vaccine!
Stay fluffy!

Synthesis Quest 1: The Blue Squeeze

LORE: A 4-month-old wrestler with a known
VSD-Leak suddenly turns blue while crying.
He starts to squat.

QUEST: Identify the Cyanotic-Secret and
explain why the Squat-Move helps his oxygen.



Synthesis Quest 2: The Projectile-Olive

LORE: A 4-week-old wrestler is throwing
milk across the room (Projectile) after
meals. He is hungry but getting smaller.
You feel a small Olive in his belly.

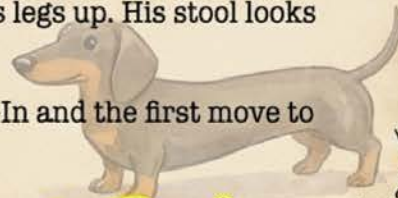
QUEST: Name the Exit-Blocker and the
Metabolic-Glitches he causes.



Synthesis Quest 3: The Currant-Jelly-Telescope

LORE: A 9-month-old wrestler has sudden fits of
screaming and drawing his legs up. His stool looks
like Red-Currant-Jelly.

QUEST: Identify the Slide-In and the first move to
Blow-the-Telescope-Back.



Doc FrizzleBob
MEDKAYFAB v1.0 COMPLETE.

